

NMAM INSTITUTE OF TECHNOLOGY, NITTE
 (An Autonomous Institution affiliated to VTU, Belagavi)
Seventh Semester B.E. (Credit System) Degree Examinations
 January 2023

19EE8X79 - ELECTRIC VEHICLE TECHNOLOGY

Duration: 3 Hours

Max. Marks: 100

Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

Unit – I		Marks	BT*	CO*
1.	a) With a neat block diagram explain a modern electric drive train system.	6	L*2	2
	b) Discuss vehicle kinetics and explain rolling resistance & aero dynamic drag.	8	L2	1
	c) Classify hybrid electric vehicles with the help of neat diagrams.	6	L2	2
2.	a) Draw and explain typical variable-speed electric motor characteristics.	6	L2	2
	b) Draw parallel hybrid electric drive train with speed coupling of planetary gear unit and explain its operation.	6	L2	2
	c) An electric vehicle has the following parameter values: $m=800$ kg, $CD=0.2$, $AF=2.2$ m ² , $CO=0.008$, $C1=1.8 \times 10^{-6}$ s ² /m ² take density of air $\rho=1.18$ kg/m ³ , and acceleration due to gravity $g=9.81$ m/s ² . The vehicle is on level road. It accelerates from 0 to 65 mph in 10 s, such that its velocity profile is given by $v(t) = 0.29055 t^2$ for $0 \leq t \leq 10$ s (i) Calculate $F_{TR}(t)$ for $0 \leq t \leq 10$ s. (ii) Calculate $P_{TR}(t)$ for $0 \leq t \leq 10$ s. (iii) Calculate the energy loss due to nonconservative forces E_{loss} . (iv) Calculate Δe_{TR} .	8	L3	1
3.	a) Compare series and parallel hybrid electric drive train with their advantages.	8	L2	2
	b) Define gradeability. Derive the equation for maximum gradeability	6	L3	1
	c) Elaborate on various operating modes in a series hybrid electric drive train system.	6	L2	2
Unit – II				
4.	a) With a neat diagram explain the components of a battery cell and list major types of rechargeable batteries considered for EV applications.	8	L2	3
	b) Explain the operation principle of a DC motor and list the four types of wound-field DC motors.	6	L2	4
	c) Explain basic fuel cell structure and plot the fuel cell characteristics.	6	L2	3
5.	a) Analyze cell discharge operation of a lead-acid battery with chemical reaction equations.	6	L3	3
	b) Explain the principle of operation of a step-down chopper with circuit diagram and waveforms	8	L2	4
	c) List major advantages of a BLDC motor.	6	L1	4
6.	a) Explain basic operation principles of Induction Motors and draw Torque–slip characteristics.	6	L2	4
	b) Define the following terms. i) Battery capacity ii) Battery discharge rate iii) Battery State of Charge	6	L1	3

P.T.O.

- c) List six major types of fuel cells and explain any two of them. 8 L2 3

Unit – III

7. a) Discuss the five operating modes in series hybrid electric drive train. 10 L2 5
b) Draw the overall control scheme of the parallel hybrid drive train and explain Engine Turn-On and Turn-Off Control Strategy used in parallel HEVs. 10 L2 5
8. a) Explain Max. SOC-of-PPS Control Strategy used in series hybrid electric drive trains. 10 L2 5
b) Discuss the parameters to be considered for engine power required at constant speed on a flat road and a 5% grade road in a parallel hybrid electric drive train. 10 L3 5

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
